Problem 22.1

$$n(T) = \int_0^\infty \frac{8\pi\nu^2}{c^3} \frac{1}{e^{\beta h\nu} - 1} d\nu$$

$$= \frac{8\pi}{c^3(\beta h)^3} \int_0^\infty \frac{x^2}{e^x - 1} dx$$

$$= \frac{8\pi}{c^3(\beta h)^3} \Gamma(3)\varsigma(3)$$

Appendix J

$$\varsigma(3) = 1.202; \varsigma(4) = \pi^4/90$$

$$\frac{k_B}{hc} = \frac{69.50356}{\mathrm{K} \cdot \mathrm{m}}$$

For $T = 2.7\mathrm{K}$

$$\frac{8\pi}{c^3(\beta h)^3}\Gamma(3)\varsigma(3) = 8\pi\left(\frac{69.5 \times 2.7}{\mathrm{m}}\right)^3 \times 2 \times 1.202$$

$$= 4.0 \times 10^8 / \mathrm{m}^3$$

$$u(T) = \int_0^\infty \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{\beta h\nu} - 1} \mathrm{d}\nu$$

$$= \frac{8\pi}{c^3 \beta (\beta h)^3} \int_0^\infty \frac{x^3}{e^x - 1} \mathrm{d}x$$

$$= \frac{8\pi}{c^3 \beta (\beta h)^3} \Gamma(4) \varsigma(4)$$

$$\frac{k_{\mathrm{B}}}{hc} = \frac{69.50356}{\mathrm{K}\cdot\mathrm{m}}$$

$$\frac{8\pi}{c^3\beta(\beta h)^3}\Gamma(4)\varsigma(4) = 8\pi hc\left(\frac{k_{\mathrm{B}}\times 2.7\mathrm{K}}{hc}\right)^4 \times 6\times\frac{\pi^4}{90}$$

$$= 8\pi(1240\mathrm{eV}\cdot\mathrm{nm})\left(\frac{69.5\times 2.7}{\mathrm{m}}\right)^4 \times 6\times\frac{\pi^4}{90}$$

$$= 2.5\times 10^5\,\mathrm{eV}/\mathrm{m}^3$$

Problem 22.4(a)

$$J = \sigma T^4$$

$$T = \left(\frac{335\,\mathrm{W/m^2}}{5.67\times 10^{-8}\,\mathrm{W/m^2K^4}} \right)^{1/4} = 2.77\times 10^2\,\mathrm{K}$$

(b)

$$335\,\mathrm{W/m^2} = J \times \frac{\pi R_\oplus^2}{4\pi R_\oplus^2} = \frac{1}{4} J$$

Problem 22.5(a)

$$T_{\max} = h\nu - \phi$$

$$= \frac{hc}{\lambda} - \phi$$

$$= \frac{1240\text{eV}\cdot\text{nm}}{200\text{nm}} - 4.1\text{eV} = 2.1\text{eV}$$

$$T_{\min} = 0$$

(b)

$$V_0 = 2.1\text{V}$$

(c)

$$\nu_C = \frac{\phi}{h}$$

$$\lambda_C = \frac{c}{\nu_C} = \frac{hc}{\phi} = \frac{1240}{4.1}\text{nm} = 302\text{nm}$$

(d)

$$I = J \cdot h\nu = J \cdot \frac{hc}{\lambda}$$

$$J = I\frac{\lambda}{hc}$$

$$= 2.0 \times \frac{200}{1240 \times 1.6 \times 10^{-19}}\text{m}^{-2}\text{s}^{-1} = 2.0 \times 10^{18}\text{m}^{-2}\text{s}^{-1}$$